

CAN Bus Data - Translated & Organized

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CAN ID (Hex)	Interval (ms)	Mode	Source Module	Message Name	Data Length	Byte Position	Parameter Description	Technical Details & Formula
0080	20	D	CON	Sync	0	-	Master Data Request / Bus Timer	-
0081	-	D	CON	Error Code	8	4+5	SEVCON Error Message	B5=High, B4=Low-Byte, B2+3=Additional data?, B6+7+8=Additional data? Example: 00 10 01 8E 48 00 00 00 = Code 0x488E = Request / 00 00 00 8E 48 00 00 08 = Error
0155	10	D+C	BMS	BMS Status 1	8	1	Charging Power Control (BMS→CHG)	Steps 7..0, Charging power = Step × 300W or Charging current = Step × 5A (at battery). Step 0 at 100% SOC
0155	10	D+C	BMS	BMS Status 1	8	2+3	Traction Battery Current	Lower 12 bits: 7D0h = 2000d zero point, Deviation × 16 = approx. Power [W], negative=discharge. Current = Deviation ÷ 4 [A]. Upper 4 bits: Charger handshake
0155	10	D+C	BMS	BMS Status 1	8	4	Phase Status	94h = Init/Exit phase = Invalid data // 54h = Valid data
0155	10	D+C	BMS	BMS Status 1	8	5+6	State of Charge (SOC)	SOC% = Decimal value ÷ 400; Only upper 13 bits used; End of charge = 9C40h
0155	10	D+C	BMS	BMS Status 1	8	7	Reserved	Fixed 0?
0155	10	D+C	BMS	BMS Status 1	8	8	Unknown Parameter	Range 5Fh - 73h

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0196	10	D	CON	Energy Status 2	8	4+5	Power Request / Torque / Slip	8 bit B4 + 1 bit B5: Rest value 254, Current power request [W] = Deviation × 250, negative = braking/reverse
0196	10	D	CON	Energy Status 2	8	6	Motor Temperature	Temperature [°C] = Value - 40
019F	10	D	CON	Energy Status 3	8	1	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	2	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	3+4	Speed (+/-)	8 bit B3 + 4 bit B4: Rest value 7D0h=2000, RPM= (Value-2000)×10, kph=RPM÷7250×80 (T80)
019F	10	D	CON	Energy Status 3	8	5	Reserved	Fixed 38h?
019F	10	D	CON	Energy Status 3	8	6	Reserved	Fixed FFh?
019F	10	D	CON	Energy Status 3	8	7	Reserved	Fixed 40h?
019F	10	D	CON	Energy Status 3	8	8	Reserved	Fixed FEh?
0423	100	D+C	CHG	Charger Status	8	1	Charger On/Off Status	03h = ON, 00h = OFF
0423	100	D+C	CHG	Charger Status	8	2	Unknown Parameter	?
0423	100	D+C	CHG	Charger Status	8	7+8	Unknown Counter	Rises to 00E0 after power-on and remains relatively constant
0424	100	D+C	BMS	BMS Status 2	8	1	Charging Protocol (BMS→CHG)	1xh = see charging protocol sheet / 9xh = Battery error
0424	100	D+C	BMS	BMS Status 2	8	2	Reserved	Fixed 40h?
0424	100	D+C	BMS	BMS Status 2	8	3	Max Regenerative Power (BMS→CON)	Current maximum regenerative power [W] = Value × 500
0424	100	D+C	BMS	BMS Status 2	8	4	Max Drive Power (BMS→CON)	Current maximum discharge power [W] = Value × 500
0424	100	D+C	BMS	BMS Status 2	8	5	Temperature (Min Cell)	Range 32h - 47h (-1/-2 diff to B8) - probably minimum cell temperature
0424	100	D+C	BMS	BMS Status 2	8	6	Battery State of Health (SOH)	In percent
0424	100	D+C	BMS	BMS Status 2	8	7	Reserved	Fixed 00h?
0424	100	D+C	BMS	BMS Status 2	8	8	Temperature (Max Cell)	Range 32h - 47h (+1/+2 diff to B5) - probably maximum cell temperature

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0425	100	D+C	BMS	BMS Status 3	8	1	Charging Protocol (BMS→CHG)	See charging protocol sheet
0425	100	D+C	BMS	BMS Status 3	8	2	Available Battery Energy	Energy amount in 1/10 kWh, includes battery aging
0425	100	D+C	BMS	BMS Status 3	8	5-6	Battery Voltage	Value = Bits 9..1 >> 1 - e.g. 310 at 57.5V
0425	100	D+C	BMS	BMS Status 3	8	7-8	Battery Voltage	Value = Bits 8..0 - e.g. 310 at 57.5V
0426	100	D+C	CHG	Vehicle Data 4	7	4-6	Odometer	km reading = Value directly [why from charger?]
0436	100	D+C	DSP	Clock	6	1-4	Minute Counter	Number of minutes since clock has power
0554	1000	D+C	BMS	Cell Temperatures	8	1-7	Cell Module Temperatures 1-7	Temperature [°C] = Value - 40
0556	100	D+C	BMS	Cell Monitor 1	8	1-8	Module Voltages 1-5	12-bit values, Voltage [V] = Decimal value ÷ 200
0557	1000	D+C	BMS	Cell Monitor 2	8	1-8	Module Voltages 6-10	12-bit values, Voltage [V] = Decimal value ÷ 200
055E	1000	D+C	BMS	Cell Monitor 3	8	1-6	Module Voltages 11-14	12-bit values, Voltage [V] = Decimal value ÷ 200
055E	1000	D+C	BMS	Cell Monitor 3	8	7-8	Battery Odometer	In 10km, e.g. 0C1Ch = 3100 = 31,000 km
055F	1000	D+C	BMS	Cell Monitor 4	8	6-8	Total Voltages	Two separate total voltages (12 bit), Voltage [V] = Decimal value ÷ 10
0597	100	D+C	CHG	12V System	8	1	System Status	See sheet 0597
0597	100	D+C	CHG	12V System	8	2	Status Flags	See sheet 0597
0597	100	D+C	CHG	12V System	8	3	DC/DC Converter Current	Rest=05, GO=07, Parking light=18, Low beam=40, High beam=45, Window heating=2B, Reverse light=15, 12V charging=03 => Assumption: Value in 1/5 A
0597	100	D+C	CHG	12V System	8	4	Charging Protocol (CHG→BMS)	Init=A1/A2/C1; ON=41h; OFF=>D1h; Charging=B1; Standby/12V maintenance=91
0597	100	D+C	CHG	12V System	8	5	Flags/Version	Fixed 29h? No, currently 2Fh (new charger)

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0597	100	D+C	CHG	12V System	8	6	Reserved	Fixed 0?
0597	100	D+C	CHG	12V System	8	7	Reserved	Fixed 1?
0597	100	D+C	CHG	12V System	8	8	Charger Temperature	Temperature [°C] = Value - 40
0599	100	D+C	DSP	Vehicle Data 1	8	1-4	Odometer	km reading = Value directly
0599	100	D+C	DSP	Vehicle Data 1	8	5	Reserved	Fixed FFh?
0599	100	D+C	DSP	Vehicle Data 1	8	6	Remaining Range	In km (decimal)
0599	100	D+C	DSP	Vehicle Data 1	8	7+8	Speed (Absolute)	Rest value 0000h .. Vmax 1FB2h = 8114 = 81.14 km/h (at display 84 km/h) (Note: 0xFFFF = invalid)
059B	100	D	CON	Drive Status 1	8	1	Gear Selection	80h=D / 20h=N / 08h=R
059B	100	D	CON	Drive Status 1	8	2	Brake Pedal/Light + Motor Status	Bit 0 (0x01) = Brake pedal/light // Bits 2+3 (0x0C) = Motor/Controller on ("ignited")
059B	100	D	CON	Drive Status 1	8	3	Power/Torque	100="Zero" / >100=Output / <100=Recuperation; Range 88...163 (?)
059B	100	D	CON	Drive Status 1	8	4	Accelerator Pedal Position	00h=Stop ... FDh=Max
059B	100	D	CON	Drive Status 1	8	5	Brake Pedal/Light + Motor Status	Upper nibble: 4=Brake inactive / 6=active // Lower nibble: 0=Motor/Controller off 4=on, 8=Error
059B	100	D	CON	Drive Status 1	8	6	Capacitor Voltage	Voltage [V] = Value ÷ 2
059B	100	D	CON	Drive Status 1	8	7	Reserved	Fixed 0?
059B	100	D	CON	Drive Status 1	8	8	Reserved	Fixed 90h = 144?
059E	100	D	CON	Vehicle Data 3	8	1+2	Distance Counter	Cyclic distance counter in 1/10 m, e.g. Value 4605 = 460.5 m, NOT trip counter!
059E	100	D	CON	Vehicle Data 3	8	6	PEM Temperature (Controller)	Temperature [°C] = Value - 40
05D7	100	D+C	DSP	Vehicle Data 2	7	1+2	Speed (Exact)	Rest value 0000h .. Vmax 1FB2h = 8114 = 81.14 km/h (at display 84 km/h)
05D7	100	D+C	DSP	Vehicle Data 2	7	3-6	Odometer	km reading = Value ÷ 1600

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05D7	100	D+C	DSP	Vehicle Data 2	7	7	Reserved	Fixed 0?
0627	100	D+C	CHG	Charger Status Message	3	1-3	Status Message CHG	Fixed 00 00 00 when BMS online
0628	100	D+C	BMS	BMS Status Message	3	1-3	Status Message BMS	Fixed 00 00 00 -- stays off when BMS offline
0629	100	D	CON	Controller Status Message	3	1-3	Status Message CON	Fixed 00 00 00 -- starts only with 59B/59E, when controller ready -- stays off when BMS offline
0659	3000	D+C	BMS	BMS Extended Status	4	1-4	Extended Status Message BMS	Fixed FF FF FF FF -- TODO: check if different in PREOP mode -- stays off when BMS offline
069F	1000	D+C	DSP	VIN	4	1-4	Serial Number	Fixed F in first nibble, other nibbles = last 7 digits of VIN backwards
0701	500	D	CON	Op/PreOp Mode	1	1	SEVCON Pre/Op Status Message	Init = 00, then 05=OP / 7F=PREOP -- stays off when BMS offline